AR Parameter estimation

$$y[t] = -a_1 y[t-1] - a_2 y[t-2] - \dots - a_p y[t-p] + f[t]$$

when $f[t]$ is
- zero-mean
- white
- variance σ_f^2

AR(p)

autoregressive

1. Mult.
2. Take expectations

$$E(y[t]\, y[t-\ell]) = E[-a_1 y[t-1] y[t-\ell] - a_2 y[t-2] y[t-\ell] - \dots - a_p y[t-p] y[t-\ell] + f[t]\, y[t-\ell]]$$

$$r_y[\ell] = -a_1 r_y[\ell-1] - a_2 r_y[\ell-2] - \dots - a_p r_y[\ell-p] + E[f[t]\, y[t-\ell]]$$

Expectations

Ex: X: throw of a die

$P(X=1) = \frac{1}{6}$

$P(X=2) = \frac{1}{6}$

$\vdots$

$P(X=6) = \frac{1}{6}$

$$E[X] = 1 \cdot P(X=1) + 2\, P(X=2) + \dots + 6\, P(X=6)$$

$$= \frac{1}{6}(1+2+3+4+5+6)$$

$$= \underline{3.5}$$

Ex: X = swap takes

$P(X = 10^6)$

$P(X = 0)$

$$E[X] = 10^6 \cdot \underbrace{P(X=10^6)}_{10^{-7}} + 0 \cdot \underbrace{P(X=0)}_{1-10^{-7}}$$

$$= 10^{-1}$$

Discrete $\quad E[X] = \sum_x x\, P(X=x)$

Cont: $\quad E[X] = \int_{-\infty}^{\infty} x\, p_X(x)\, dx$

- Expectations are linear operators.

$$E[a_1 X_1 + a_2 X_2 + \dots + a_n X_n] = E[a_1 X_1] + E[a_2 X_2] + \dots + E[a_n X_n]$$

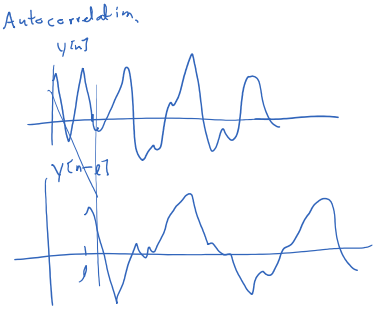
- Non random factors can be pulled out of expectations.

$$= a_1 E[X_1] + a_2 E[X_2] + \dots + a_n E[X_n]$$

$y[n]$: random process.

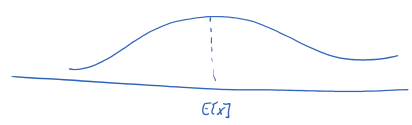
$E[y[n]]$ = mean.

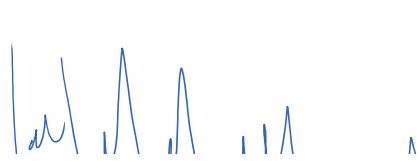
Autocorrelation.

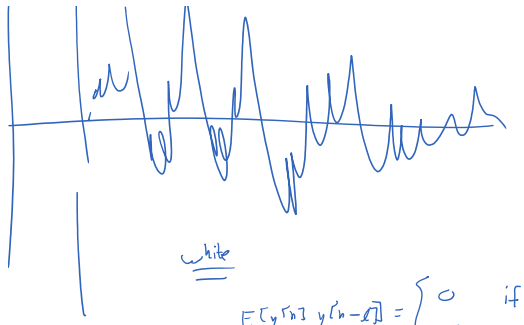


Mean: $E[X]$

Variance: $\sigma^2 = E[(X - E[X])^2] = E[X^2] - (E[X])^2$







$r_y(\ell) = r(\ell) = E[\, y[n]\, y[n-\ell] \,]$

$r_{yy}(\ell)$

white

$$E[y[n]\, y[n-\ell]] = \begin{cases} 0 & \text{if } \ell \neq 0 \\ \sigma_y^2 & \text{if } \ell = 0 \end{cases}$$

$$\boxed{r_y[\ell] = -a_1\, r_y[\ell-1] - \cdots - a_p\, r_y[\ell-p]} + \underbrace{E[\, y[t-\ell]\, f[t] \,]}_{\cancel{0} \text{ if } \ell > 0}$$

$$\begin{array}{c} \ell=1 \\ \ell=2 \\ \\ \ell=p \end{array} \underbrace{\begin{bmatrix} r_y[0] & r_y[-1] & \cdots & r_y[1-p] \\ r_y[1] & r_y[0] & \cdots & r_y[2-p] \\ \vdots & & & \\ r_y[p-1] & r_y[p-2] & \cdots & r_y[0] \end{bmatrix}}_{R} \underbrace{\begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_p \end{bmatrix}}_{(\,q\,)} = \underbrace{\begin{bmatrix} r_y[1] \\ r_y[2] \\ \vdots \\ r_y[p] \end{bmatrix}}_{p}$$

Toeplitz matrix

Yule-Walker

Real: $R = R^T$ (symmetric)

Complex: $R = R^H$

H: conjugate transpose.

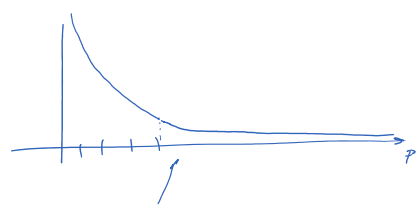
$R\, q = p. \qquad q = R^{-1} p.$

$[\, r[0] \,]$

$\begin{bmatrix} r[0] & r[-1] \\ r[1] & r[0] \end{bmatrix}$

$\begin{bmatrix} r[0] & r[-1] & r[-2] \\ r[1] & r[0] & r[-1] \\ r[2] & r[1] & r[0] \end{bmatrix}$

$E[(y[n] - \hat{y}_p[n])^2]$



$(n+1) \times (n+1)$:

$$T_n = \begin{bmatrix} r_0 & r_1 & \cdots & r_n \\ r_1 & r_0 & \cdots & r_{n-1} \\ \vdots & & & \\ r_n & & \cdots & r_0 \end{bmatrix} \qquad T_0 = r_0 \qquad T_1 = \begin{bmatrix} r_0 & r_1 \\ r_1 & r_0 \end{bmatrix}, \text{ etc.}$$

$$T_n \underbrace{\begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_{n+1} \end{bmatrix}}_{\underline{w}_n} = \underbrace{\begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_{n+1} \end{bmatrix}}_{\underline{r}_n}$$

Fact: Let $J = \begin{bmatrix} 0 & & & 1 \\ & & 1 & \\ & 1 & & \\ 1 & & & 0 \end{bmatrix}$

If T is a Toeplitz matrix then $T = J\, T^T J$

$[x_1] \quad [x_n]$

If T is a <u>Toeplitz</u> matrix then $\underline{\underline{T = \underbrace{J T^T J}_{\text{Persymmetric}}}}$

pf: $J \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_n \\ x_{n-1} \\ \vdots \\ x_1 \end{bmatrix}$

$[x_1 \cdots x_n] J = [x_n \; x_{n-1} \cdots x_1]$

$$J T^T = J \begin{bmatrix} r_0 & r_1 & \cdots & r_n \\ \vdots & & & \\ r_{n-1} & r_{n-2} & & \\ r_n & r_{n-1} & & r_0 \end{bmatrix} = \begin{bmatrix} r_n & r_{n-1} & & r_0 \\ r_{n-1} & r_{n-2} & \cdots & \vdots \\ \vdots & \vdots & & \\ r_0 & r_1 & & r_n \end{bmatrix}$$

$$J T^T J = \begin{bmatrix} r_0 & r_1 & \cdots & r_n \\ r_1 & r_0 & \cdots & r_{n-1} \\ \vdots & & & \\ r_n & \cdots & & r_0 \end{bmatrix}$$

$$T = J T^T J$$

$$\begin{cases} J T = J^2 T^T J \\ \quad = T^T J \end{cases}$$

$$(JT)^{-1} = (T^T J)^{-1}$$

$$= T^{-1} J^{-1} = J^{-1} T^{-T}$$

$$= T^{-1} J = J T^{-T}$$

$$TJ = J T^T J^2 = J T^T$$

$$(TJ)^{-1} = (J T^T)^{-1}$$

$$\underline{J T^{-1} = T^{-T} J}$$

Inductively, suppose we have the solution $T_n \underline{w}_n = \underline{r}_n$

want to do: use this to find the solution to $T_{n+1} \underline{w}_{n+1} = \underline{r}_{n+1}$

$$\underline{w}_{n+1} = \begin{bmatrix} \underline{z}_n \\ \alpha_n \end{bmatrix}$$

$$T_{n+1} = \left[\begin{array}{c|c} T_n & \begin{matrix} r_{n+1} \\ r_n \\ \vdots \\ r_1 \end{matrix} \\ \hline r_{n+1} \; r_n \cdots r_1 & r_0 \end{array}\right] = \left[\begin{array}{c|c} T_n & J_n \underline{r}_n \\ \hline \underline{r}_n^T J_n & r_0 \end{array}\right]$$

$$T_{n+1} \underline{w}_{n+1} = \underline{r}_{n+1}$$

$$\left[\begin{array}{c|c} T_n & J_n \underline{r}_n \\ \hline \underline{r}_n^T J_n & r_0 \end{array}\right] \begin{bmatrix} \underline{z}_n \\ \alpha_n \end{bmatrix} = \begin{bmatrix} \underline{r}_n \\ r_{n+2} \end{bmatrix}$$

$$\begin{cases} T_n \underline{z}_n + J_n \underline{r}_n \alpha_n = \underline{r}_n \\ \underline{r}_n^T J_n \underline{z}_n + r_0 \alpha_n = r_{n+2} \end{cases} \longrightarrow \underline{z}_n = T_n^{-1} (\underline{r}_n - J_n \underline{r}_n \alpha_n)$$

$$= \underline{w}_n - T_n^{-1} J_n \underline{r}_n \alpha_n$$

$$= \underline{w}_n - J_n T_n^{-T} \underline{r}_n \alpha_n$$

$$= \underline{w}_n - J_n \underline{w}_n \alpha_n$$

$$\underline{r}_n^T J_n (\underline{w}_n - J_n \underline{w}_n \alpha_n) + r_0 \alpha_n = r_{n+2}$$

$$(-\underline{r}_n^T \underline{w}_n + r_0) \alpha_n = r_{n+2} - \underline{r}_n^T J_n \underline{w}_n$$

$$\boxed{\alpha_n = \frac{r_{n+2} - \underline{r}_n^T J_n \underline{w}_n}{r_0 - \underline{r}_n^T \underline{w}_n}}$$

In Matlab, ' is conjugate transpose.

.' is transpose.

$$E\Big[\underbrace{\big(y[n] + a_1 y[n-1] + \dots + a_p y[n-p] - f[t]\big)}_{\text{error}} \underbrace{y[n-l]}_{\text{"data"}}\Big] = 0$$

$$\underline{y}[n-1] = \begin{bmatrix} y[n-1] \\ y[n-2] \\ \vdots \\ y[n-p] \end{bmatrix} \qquad \underline{a} = \begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_p \end{bmatrix}$$

$$\hat{y}[n] = -a_1^* y[n-1] - \dots - a_p^* y[n-p]$$
$$= \underline{a}^H \underline{y}[n-1]$$

choose $\{a_1, a_2, \dots, a_p\}$ so that $E\left[\left(y[n] - \hat{y}[n]\right)^2\right]$ is small.

$$\downarrow$$

$$\min_{\{a_1, \dots, a_p\}} E\left[\left(y[n] - \underline{a}^T \underline{y}[n-1]\right)^2\right]$$

$$E\left[y[n]^2 - 2\, y[n]\, \underline{a}^T \underline{y}[n-1] + \underline{a}^T \underline{y}[n-1]\, \underline{y}^T[n-1]\, \underline{a}\right]$$

$$= E[y[n]^2] - 2\underline{a}^T \underbrace{E[y[n]\underline{y}[n-1]]}_{\underline{p}} + \underline{a}^T \underbrace{E[\underline{y}[n-1]\underline{y}^T[n-1]]}_{R}\, \underline{a}$$

$$= E[y[n]^2] - 2\underline{a}^T \underline{p} + \underline{a}^T R\, \underline{a}$$

Linear

scalar

$y = ax$

$$\frac{\partial y}{\partial a} = \frac{\partial ax}{\partial a} = x$$

Real Vector

$y = \underline{a}^T \underline{x}$

$$\frac{\partial y}{\partial \underline{a}} = \frac{\partial(\underline{a}^T \underline{x})}{\partial \underline{a}} = \begin{bmatrix} \frac{\partial(\underline{a}^T\underline{x})}{\partial a_1} \\ \frac{\partial(\underline{a}^T\underline{x})}{\partial a_2} \\ \vdots \\ \frac{\partial(\underline{a}^T\underline{x})}{\partial a_n} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \underline{x}$$

$$\boxed{\frac{\partial(\underline{a}^T\underline{x})}{\partial \underline{a}} = \underline{x}}$$

Complex

Wirtinger Calculus

z : complex

$z = x + jy$

$$\boxed{\frac{\partial z}{\partial z} = 1 \qquad \frac{\partial z^*}{\partial z} = 0}$$

$$\boxed{\frac{\partial z}{\partial z^*} = 0}$$

$y = \underline{a}^H \underline{x}$

$$\boxed{\frac{\partial(\underline{a}^H\underline{x})}{\partial \underline{a}^*} = \underline{x}} \qquad \frac{\partial(\underline{a}^T\underline{x})}{\partial \underline{a}^*} = \underline{0}$$

$$\boxed{\frac{\partial(\underline{a}^T\underline{x})}{\partial \underline{a}} = \underline{x}}$$

$$\boxed{\frac{\partial(\underline{a}^H\underline{x})}{\partial \underline{a}^*} = \underline{x}} \qquad \frac{\partial(\underline{a}^T\underline{x})}{\partial \underline{a}^*} = \underline{0}$$

$$\frac{\partial\, \underline{x}^H\underline{a}}{\partial \underline{a}} = \underline{x}^* \qquad \frac{\partial\, \underline{x}^T\underline{a}^*}{\partial \underline{a}} = \underline{0}$$

Quadriet

$y = a^2 r$

$$\frac{\partial y}{\partial a} = \frac{\partial(a^2 r)}{\partial a} = 2ar$$

$y = \underline{a}^T R\, \underline{a}$, R symmetric, $R = R^T$

$$\boxed{\frac{\partial(\underline{a}^T R\, \underline{a})}{\partial \underline{a}} = 2R\,\underline{a}}$$

$y = \underline{a}^H R\, \underline{a}$ $\quad R = R^H$

$$\boxed{\frac{\partial(\underline{a}^H R\, \underline{a})}{\partial \underline{a}^*} = R\,\underline{a}}$$

Q: Is y real?

$y^H = (\underline{a}^H R\, \underline{a})^H$

$= \underline{a}^H R^H \underline{a} = y$

$E[(y[n] - \hat{y}[n])^2] = E[y[n]^2] - 2\underline{a}^T\underline{p} + \underline{a}^T R\, \underline{a}$

$E[|y[n] - \hat{y}[n]|^2] = E[|y[n]|^2] - 2\,\mathrm{Re}\,\underline{a}^H\underline{p} + \underline{a}^H R\, \underline{a}$

$\frac{\partial}{\partial \underline{a}} E[(y[n] - \hat{y}[n])^2] = -2\underline{p} + 2R\underline{a}$

$\Rightarrow \boxed{R\underline{a} = \underline{p}} \leftarrow$ Yule-Walker

$\frac{\partial}{\partial \underline{a}^*} E[|y[n] - \hat{y}[n]|^2] = -\underline{p} + R\underline{a}$

$\frac{\partial}{\partial \underline{a}}(\quad) = -\underline{p}^* + R^T\underline{a}^*$

$z = x + jy$

$z^* = x - jy$

$\frac{z + z^*}{2} = x = \mathrm{Re}[z]$

$2\,\mathrm{Re}(\underline{a}^H\underline{p}) = (2)\left(\frac{1}{2}\right)(\underline{a}^H\underline{p} + \underline{a}^T\underline{p}^*)$

$$y[n] = -\sum_{i=1}^{p} a_i\, y[n-i] + f[n]$$

$y[1], y[2], \ldots, y[N]$

$$\underbrace{\begin{bmatrix} y[p] & y[p-1] & \cdots & y[2] & y[1] \\ y[p+1] & y[p] & \cdots & y[3] & y[2] \\ & & \vdots & & \\ y[N-1] & y[N-2] & \cdots & & y[N-p] \end{bmatrix}}_{Y} \underbrace{\begin{bmatrix} -a_1 \\ -a_2 \\ \vdots \\ -a_p \end{bmatrix}}_{\underline{w}} = \underbrace{\begin{bmatrix} y[p+1] \\ y[p+2] \\ \vdots \\ y[N] \end{bmatrix}}_{\underline{y}} + \underbrace{\begin{bmatrix} -f[p+1] \\ -f[p+2] \\ \vdots \\ f[N] \end{bmatrix}}_{\underline{f}}$$

$$\boxed{Y\underline{w} = \underline{y} + \underline{f}}$$

Maximum likelihood estimation

$\underline{f} \sim N(\underline{0}, Q)$.

$\underline{y} = A\underline{w} + \underline{f}$, A is known matrix

$\underline{y} \mid A\underline{w}$

$\underline{y} \mid \underline{w}$ $\quad \sim N(Aw, Q)$

$\underline{N} \sim N(\underline{\mu}, Q)$, $\quad \dim(\underline{N}) = d$

$f(n) = \frac{1}{\quad} \exp\left[-\frac{1}{\quad}(n-\mu)^T Q^{-1}(n-\mu)\right]$

$n \sim N(0, \sigma^2)$ $\qquad f(n) = \frac{1}{\sqrt{2\pi}\sigma} e^{-n^2/2\sigma^2}$

$y = 3 + n$

$y \sim N(3, \sigma^2)$

$y = s + n$, $\quad n \sim N(0, \sigma^2)$

s: r.v.

$y \mid s \sim N(s, \sigma^2)$

"Given s"

$\underline{N} \sim \mathcal{N}(\underline{\mu}, Q)$, $\quad \dim(\underline{N}) = d$.

"Given y"

$$f_{\underline{N}}(\underline{n}) = \frac{1}{(2\pi)^{d/2} \, |\det(Q)|^{1/2}} \exp\left[-\frac{1}{2}(\underline{n} - \underline{\mu})^T Q^{-1} (\underline{n} - \underline{\mu})\right]$$

$$f_{\underline{Y}}(\underline{y} \mid \underline{w}) = \frac{1}{(2\pi)^{d/2} \, |\det(Q)|^{1/2}} \exp\left[-\frac{1}{2}(\underline{y} - A\underline{w})^T Q^{-1} (\underline{y} - A\underline{w})\right]$$